

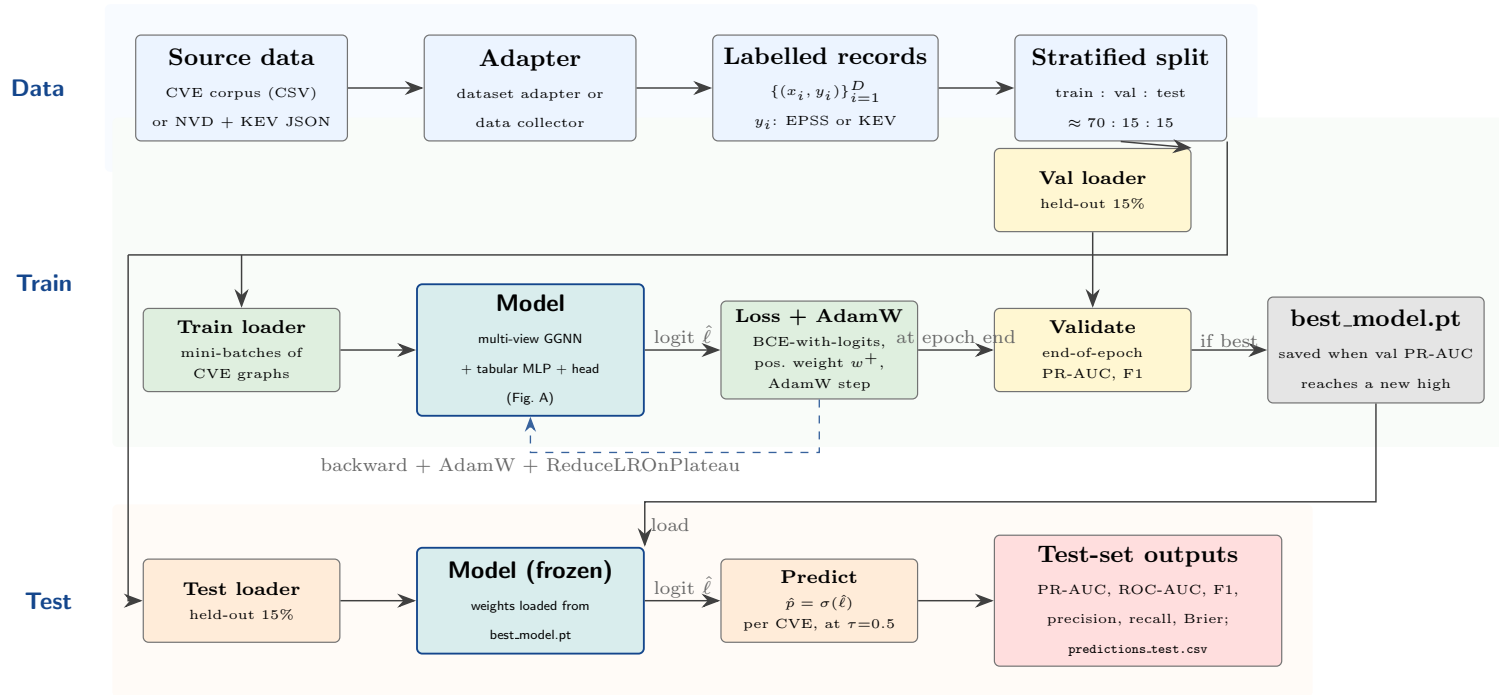


Figure. How a single per-CVE model (Figure A) is used across the full training and testing lifecycle. *Data row:* every CVE record is turned by the dataset adapter into a labelled pair (x_i, y_i) , and the full set $\{(x_i, y_i)\}_{i=1}^D$ is stratified-split into train, validation and test subsets of roughly 70/15/15. *Train row:* each mini-batch from the train loader is processed by the model, the logit $\hat{\ell}$ is turned into a binary cross-entropy loss with a positive-class weight w^+ equal to the negative-to-positive ratio of the train split, and AdamW updates the weights with ReduceLROnPlateau decay. At the end of every epoch the same model runs over the validation loader; whenever PR-AUC reaches a new high the weights are saved as `best_model.pt`, and training stops early after 15 epochs without improvement. *Test row:* the saved weights are loaded into a frozen forward pass over the test loader, the sigmoid $\sigma(\cdot)$ converts the logit $\hat{\ell}$ into the predicted exploitation probability $\hat{p} = \sigma(\hat{\ell})$, and PR-AUC, ROC-AUC, F1, precision, recall and Brier are computed at the default decision threshold $\tau = 0.5$.